Machine Translation for Lingala–French and Tigrinya–English

Fine-Tuning mBART-50 on Two Low-Resourced African Language Pairs

AIMS | African Institute for Mathematical Sciences CAMEROON

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Outline

1. Motivation
2. Languages & Data
3. Model & Setup
4. Results
5. Discussion
6. Conclusion

Motivation

The Problem

Of 7,000+ living languages, fewer than **100** receive NLP attention. Africa has **2,000+** languages — most absent from MT benchmarks.

- mBART-50 covers 50 languages: **Lingala** and **Tigrinya** are not included
- **Fake language code** strategy (MAFAND-MT): map an unsupported language to a related supported slot

Our Goal

Provide the **first mBART-50 baselines** for Lingala–French and Tigrinya–English, and assess the limits of the fake code strategy.

Languages at a Glance

Lingala (Bantu)

Family	Niger-Congo / Bantu
Script	**Latin**
Proxy	`sw_KE` (Swahili)
Target	French (`fr_XX`)

Shares script & family with proxy ⇒ strong fit

Tigrinya (Ethiosemitic)

Family	Afroasiatic
Script	**Ge'ez (Ethiopic)**
Proxy	`sw_KE` (Swahili)
Target	English (`en_XX`)

Different script & family ⇒ weak fit

Dataset

5,000 parallel pairs per language pair | Split: Train 4,000 / Val 500 / Test 500
Lingala–Fr: general domain | Tigrinya–En: Gamayun project (conversational)

Model & Fine-Tuning Setup

mBART-50 (610M params)

- Encoder-decoder Transformer, pre-trained on 50 languages
- Both pairs mapped to `sw_KE` proxy slot
- `forced_bos_token` steers decoder to correct target

Evaluation Metrics

- **BLEU** — n-gram precision
- **ChrF** — character-level F-score
- **TER** — edit rate (*lower = better*)
- **METEOR** — semantic overlap

Key Hyperparameters

Epochs / Batch size	5 / 8
Learning rate	3×10^{-5}
Beam size / Max length	4 / 128 tokens
Precision	fp16 (mixed)

Fake Code Condition

✓ Works: same script & family

× Fails: different script & family

Main Results — Corpus-Level Metrics

Language Pair	BLEU ↑	ChrF ↑	TER ↓	METEOR ↑
Lingala–French	**15.09**	**35.42**	**78.71**	**38.37**
Tigrinya–English	3.52	20.46	99.11	20.25

Lingala–French

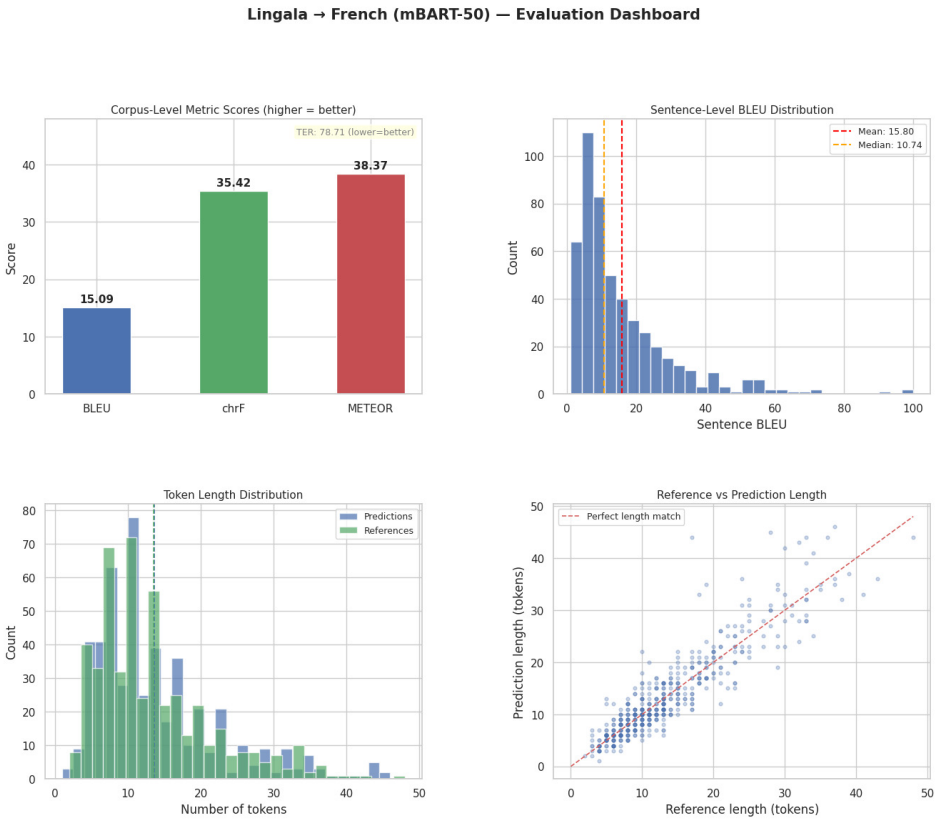
Competitive with MAFAND-MT Bantu baselines
(cf. Hausa–English: BLEU 16.0, 3,098 pairs)

Tigrinya–English

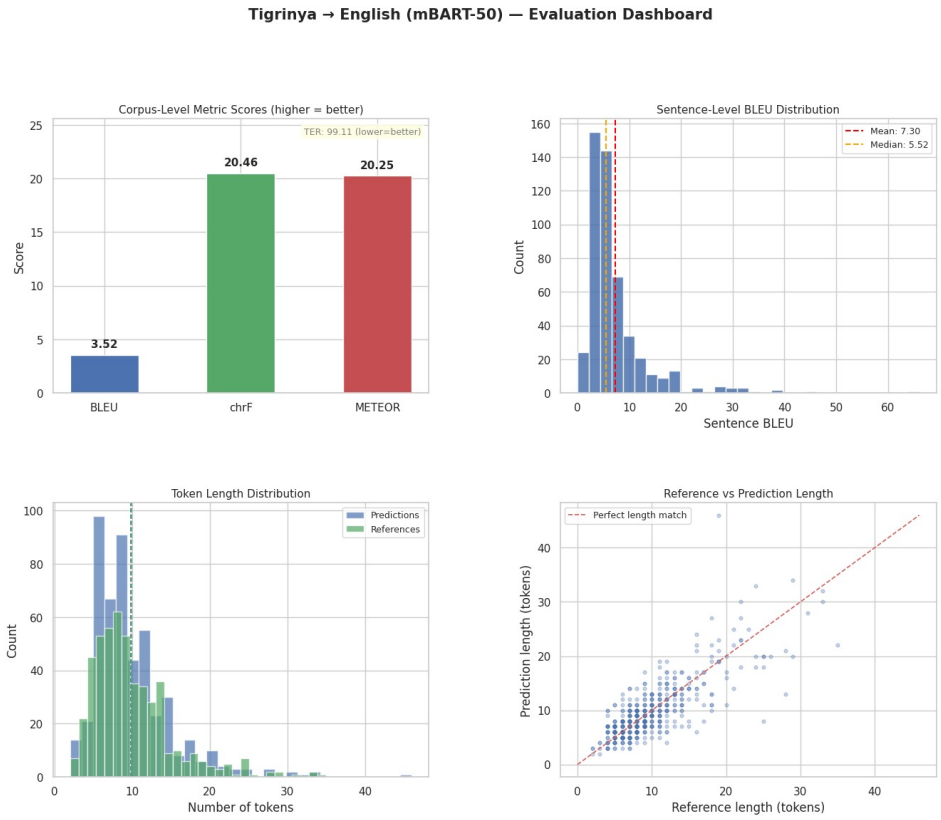
TER ≈ 99: output needs *nearly as many edits* as the reference — near-random output

Evaluation Dashboards

Lingala → French (mBART-50)



Tigrinya → English (mBART-50)



Comparison with MAFAND-MT Baselines

System	Language Pair	Train Pairs	BLEU	ChrF
Our work	Lingala–French	4,000	15.09	35.42
Our work	Tigrinya–English	4,000	3.52	20.46
Adelani et al.	Bambara–French	3,302	22.7	48.2
Adelani et al.	Hausa–English	3,098	16.0	43.3
Adelani et al.	Igbo–English	6,998	24.7	50.0
Adelani et al.	Swahili–English	30,782	26.7	56.4

Lingala–French is within the MAFAND-MT range. Tigrinya–English lags well below all baselines — script mismatch is the bottleneck.

Qualitative Translation Examples

Lingala–French: Best (BLEU 100)

Source	*Bozalisi elango kiziba?*
Reference	Est-ce que la porte était fermée ?
Prediction	Est-ce que la porte était fermée ? ✓ exact match

Lingala–French: Worst (BLEU 1.10) — Hallucination

Reference: *Les boissons alcoolisées... Le risque augmente avec la dose.*
Prediction: *Les soins dentaires...* [unrelated medical text, repetitive]

Tigrinya–English: Best (BLEU 66.06)

Ref: *...do that* **this week**.
Pred: *...do that* **today**. minor shift

Tigrinya–English: Worst (BLEU 0)

Ref: *Have you considered becoming a lawyer?*
Pred: *You can't accuse him of stealing.*

Why Does Lingala Outperform Tigrinya?

Script

Lingala: **Latin** ⇒ shared with French & Swahili proxy.

Tigrinya: **Ge'ez** ⇒ nearly absent from mBART-50 vocabulary.

Linguistic Proximity

Lingala & Swahili: both **Bantu** ⇒ similar morphology & word order.

Tigrinya: **Afroasiatic** ⇒ remote from proxy in all dimensions.

Decoder Priors

French is **heavily represented** in mBART-50 pre-training.

Broken Ge'ez encoding forces English decoder to hallucinate.

Key Finding

Fake code effectiveness scales with **script compatibility** and **linguistic proximity** between source and proxy.

Paths Forward for Tigrinya

1. Byte-level models (ByT5)

No vocabulary needed for Ge'ez; shown to outperform mT5 by 3+ BLEU on African languages.

2. Dedicated Ge'ez tokenisation

Custom SentencePiece vocabulary + mBART-50 embedding extension.

3. Larger parallel corpus

Quality gains scale to ∼2,500 in-domain pairs; a Tigrinya news corpus would help.

4. Continual pre-training

Pre-train on Tigrinya monolingual data (Bible, JW300, web) before fine-tuning.

Summary & Limitations

Key Takeaways

- First mBART-50 baselines: Lingala–Fr (**BLEU 15.09**) and Tigrinya–En (**BLEU 3.52**)
- Fake code **works** for Latin-script Bantu; **fails** for Ge'ez Ethiosemitic
- Lingala–French competitive with MAFAND-MT Bantu pairs
- ByT5 / byte-level models are the best path for Tigrinya

Limitations

- Only 4,000 training pairs per language pair
- Automatic metrics only; no human evaluation
- Single model & configuration tested

Ethics

Data under licence; no PII; code & data released under **CC-BY 4.0**

Thank you for your attention!

Questions & Discussion

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